

quantization of Hamiltonian actions

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Now we may start with the orbit method/geometric quantization

Intuition

We want the following correspondence

symplectic geometry	representation theory
Hamiltonian $G \curvearrowright M$	representation $G \curvearrowright \mathcal{H}(M)$

symplectic geometry	representation theory
moment map $\begin{array}{c} M \\ \downarrow \mu \\ \mathfrak{g}^* \\ \coprod_{\alpha} \mathcal{O}_{\alpha} \end{array}$	decomposition into irreps $\begin{array}{c} \mathcal{H} \\ \downarrow \\ \bigoplus_{\alpha} E_{\alpha} \end{array}$
$\{f, g\}$	$\{A, B\}$
	?
orbit $\mathcal{O}_{\alpha}$	irrep E_{α}
symplectic reduction $M//G = \mu^{-1}(0)/G$	$\mathcal{H}(M//G) = \mathcal{H}(M)^G$
Hamiltonian restriction	restricted representation
symplectic induction	induced representation
	Peter-Weyl theorem

- Borel-Weil-Bott theorem
- orbit method
 - Compact Lie groups
- Nilpotent Lie groups

- On induced representations
-

- <https://chessapig.github.io/talk/G-spaces>
- <https://www.numdam.org/item/10.5802/ccirm.4.pdf>

motivating Hamiltonian flows

on the plane

We want a vector field that *preserves* a given smooth

$$H : \mathbb{R}^2 \rightarrow \mathbb{R}$$

Then

$$\text{grad}(H)$$

does the opposite: its flow goes out of $H^{-1}(c)$ *curves*.

So we should work with the 90 degrees rotation of the gradient, so

$$J\text{grad}(H)$$

where

$$J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

is the clockwise 90 degrees rotation matrix.

The flow of this vector field ϕ^t such that

$$\frac{d}{dt}\phi^t(p) = J\text{grad}(H)_{\phi^t(p)}$$

which quickly implies

$$\frac{d}{dt}H(\phi^t(p)) = dH\left(\frac{d}{dt}\phi^t(p)\right) = \langle \text{grad}(H), J\text{grad}(H) \rangle = 0$$

by chain rule!

Hamiltonian vector fields on plane preserve area

$$X_H = \begin{bmatrix} -\frac{\partial H}{\partial y} \\ \frac{\partial H}{\partial x} \end{bmatrix}$$

$$\text{div}(X_H) = -\frac{\partial}{\partial x} \frac{\partial H}{\partial y} + \frac{\partial}{\partial y} \frac{\partial H}{\partial x} = 0$$

And divergence free vector fields preserve area (volume in $\mathbb{R}^n$)

are all area preserving flows Hamiltonian vector fields?

Well,

- $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ is *Hamiltonian* for some smooth H
- $\Longleftrightarrow$

$$\exists H \in C^\infty(\mathbb{R}^2) : \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} -\frac{\partial H}{\partial y} \\ \frac{\partial H}{\partial x} \end{bmatrix}$$

- $\iff$

$$\exists H \in \mathcal{C}^\infty(\mathbb{R}^2) : dH = X_2 dx - X_1 dy$$

- So we may use facts about 1-forms

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \leftrightarrow \alpha = X_2 dx - X_1 dy$$

- We know **on a simply connected domain** α is closed ($d\alpha = 0$) $\iff \alpha$ is exact

$$\exists H \in \mathcal{C}^\infty(\mathbb{R}^2) : \alpha = dH.$$

- Now

$$d\alpha = \underbrace{\left(-\frac{\partial X_1}{\partial x} - \frac{\partial X_2}{\partial y} \right)}_{-\text{div}(X)} dx \wedge dy$$

- Hence, on a **simply connected domain** flow of X is area preserving $\iff X$ is divergence free $\iff$

$$\exists H \in \mathcal{C}^\infty(\mathbb{R}^2) \text{ such that}$$

$$X = J \text{grad}(H)$$

-  Notice a minus sign pops out in $d\alpha$ above.

a area preserving flow that is *not* globally Hamiltonian on the punctured plane

The 1-form

📌 **Definition. Angle form on $\mathbb{R}^2 \setminus \{0\}$**

The angle form on $\mathbb{R}^2 \setminus \{0\}$ is defined by

$$\frac{-ydx + xdy}{x^2 + y^2}$$

using the coordinates (x, y) .

is closed but not exact.

So equating this form with $-\alpha$ above we get the vector field

$$X_\theta := \frac{1}{x^2 + y^2} \begin{bmatrix} x \\ y \end{bmatrix}$$

which has zero divergence but does not have a globally defined Hamiltonian.

It is however locally Hamiltonian for the angle function

$$X_\theta = J \text{grad}(\theta)$$

defined on any simply connected subset of $\mathbb{R}^2 \setminus \{0\}$.

Hamiltonian but in different coordinates

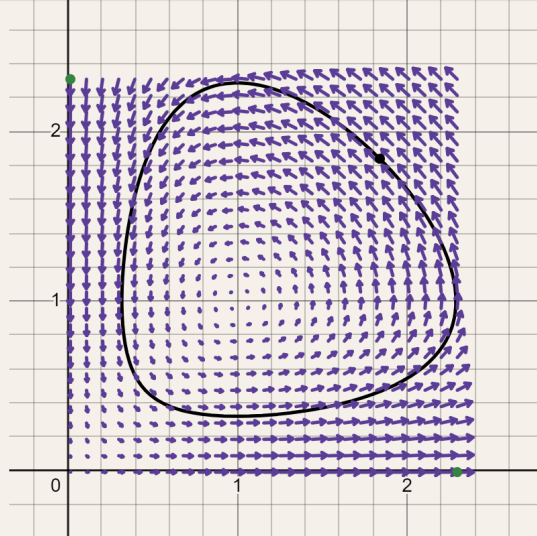
space.f.field.usual Lotka-Volterra

Lotka-Volterra vector field on $\mathbb{R}^2$

 **Definition.** Lotka-Volterra vector field on $\mathbb{R}^2$

$$\begin{bmatrix} \lambda_1 x - \eta_1 xy \\ -\lambda_2 y + \eta_2 xy \end{bmatrix}$$

which looks like



on $\mathbb{R}_{>0}^2$
[1]

for $\lambda_i, \eta_i = 1$

- $$\operatorname{div} \begin{bmatrix} x - xy \\ -y + xy \end{bmatrix} = 1 - y - 1 + x = x - y$$

The flow is not area preserving of course, but it has a "*hidden area preserving*" because the flow is periodic on $\mathbb{R}_{>0}^2$.

- However, it preserves a *different* area form on $\mathbb{R}_{>0}^2$

$$\mathfrak{L}_X \left(\frac{dx \wedge dy}{xy} \right) = \operatorname{div} \left(\frac{1}{xy} X \right) = 0$$

- Consider the smooth function for $(x, y) \in \mathbb{R}_{>0}^2$

$$x - \log x + y - \log y$$

- The vector field preserves this function

$$\begin{aligned} \partial_{\begin{bmatrix} x-xy \\ -y+xy \end{bmatrix}} (x - \log x + y - \log y) &= (x - xy) \left(1 - \frac{1}{x} \right) + (-y + xy) \left(1 - \frac{1}{y} \right) \\ &= (x - xy - y + xy) - 1 + y + 1 - x \\ &= 0 \end{aligned}$$

- However,

$$\begin{aligned} \operatorname{grad}(x - \log x + y - \log y) &= \begin{bmatrix} 1 - \frac{1}{x} \\ 1 - \frac{1}{y} \end{bmatrix} \\ \implies J \operatorname{grad}(x - \log x + y - \log y) &= \begin{bmatrix} -1 + \frac{1}{y} \\ 1 - \frac{1}{x} \end{bmatrix} = \frac{1}{xy} \begin{bmatrix} x - xy \\ -y + xy \end{bmatrix} \end{aligned}$$

- This shows the vector field is Hamiltonian vector field with the symplectic form $\frac{1}{xy}dx \wedge dy$ on $\mathbb{R}_{>0}^2$.

✓✓ <https://www.desmos.com/calculator/l0gvaeds5s>

✓✓ [Lotka-Volter flow](#)

in logarithmic coordinates

- In coordinates

$$\begin{aligned} \mathbb{R}_{>0}^2 &\rightarrow \mathbb{R}_{>0}^2 \\ \begin{pmatrix} x \\ y \end{pmatrix} &\mapsto \begin{pmatrix} \log x \\ \log y \end{pmatrix} \end{aligned}$$

we have

Now it is a Hamiltonian system.

1. <https://www.desmos.com/calculator/l0gvaeds5s> ↩

☰ Todo

- Lotka-Volterra's vector field
- diagonal action on $\mathbb{C}^n$

sidenote: vector fields that are (locally) Hamiltonian as well as (locally) gradient

<i>object</i>	equivalent to Cauchy-Riemann operator	Cauchy-Riemann operator/equivalent gives 0 which makes the object a	if Cauchy-Riemann operator/equivalent gives 0 then a primitive exists <i>locally</i>
a complex function $f = u + iv$	the Cauchy-Riemann operator $\frac{\partial}{\partial \bar{z}} f = \frac{1}{2} \left((u_x - v_y) + i(v_x + u_y) \right)$	Holomorphic function	$\exists g$ $\frac{\partial g}{\partial \bar{z}} = f$
a (complex) differential (1,0)-form $f(z)dz$	the exterior derivative $d(f(z)dz) = -\frac{\partial f}{\partial \bar{z}} dz \wedge d\bar{z}$	closed differential 1-form/Holomorphic 1-form	$\exists g dz$ $dg = f dz$

<i>object</i>	equivalent to Cauchy-Riemann operator	Cauchy-Riemann operator/equivalent gives 0 which makes the object a	if Cauchy-Riemann operator/equivalent gives 0 then a primitive exists <i>locally</i>
(real) differential 1-form $\omega = udx - vdy$ and its Hodge dual $\star\omega = vdx + udy$	$d\omega = (u_y + v_x)dx \wedge dy$ $d\star\omega = (v_y - u_x)dx \wedge dy$	closed, coclosed (real) differential 1-form	$\exists \alpha, \beta$ (real) 1-forms $d\alpha = \omega, d\beta = \star\omega$
(real) vector field $\bar{f} = \begin{bmatrix} u \\ -v \end{bmatrix}$	$\text{curl} \bar{f} = -v_x - u_y$ $\text{div} \bar{f} = u_x - v_y$	incompressible (area preserving/symplectic) and a solenoid (curl free) vector field	$\exists H, \phi$ such that $J\text{grad}(H) = \bar{f} = \text{grad}(\phi)$

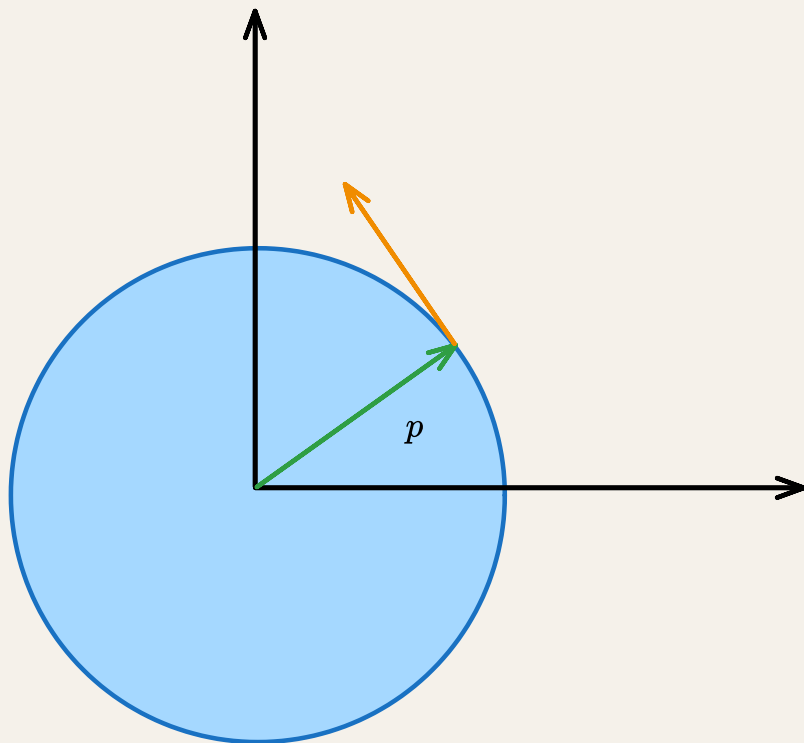
on the 2-sphere

Almost complex structure on S^2 induced from spherical metric and its area form

 **Definition.** Almost complex structure on S^2 using cross product on $\mathbb{R}^3$.

The involution

$$J : TS^2 \rightarrow TS^2$$
$$X_p \mapsto p \times X_p$$



where we've identified $T_p S^2 \leq \mathbb{R}^3$.

Because area spanned by the parallelogram is

$$\|X\| \|Y\| \sin(\theta_{X,Y})$$

and *oriented area* is

- supposed to be

$$\begin{aligned}\langle X, Y \rangle &= \|X\| \|Y\| \cos(\theta_{X,Y}) \\ &= \|X\| \|Y\| \sin\left(\theta_{X,Y} + \frac{\pi}{2}\right) \\ &= \langle JX, Y \rangle\end{aligned}$$

- $$\langle JX, Y \rangle = \langle p \times X, Y \rangle = \langle p, X \times Y \rangle$$

which, again, *should* be the oriented area of $X, Y \in T_p S^2$

- In local coordinates

$$\langle JX, Y \rangle = \left\langle J \frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right\rangle dx \wedge dy = \left\| \frac{\partial}{\partial x} \times \frac{\partial}{\partial y} \right\| dx \wedge dy$$

so we conclude

$$g_S(JX, Y) = \lambda_S(X, Y)$$

So same as before we define

🔑 **Definition.** Hamiltonian vector field of a smooth function on the 2-sphere $H \in \mathcal{C}^\infty(S^2)$ is the vector field

$$X_H = J\text{grad}(H)$$

- It preserves H because of chain rule, again.
- It preserves area

on complex vector spaces

summary of Hamiltonian vector fields in different spaces

$\mathbb{C} = \mathbb{R}^2$	S^2	$\mathbb{C}^n = \mathbb{R}^{2n}$	(M, ω) <i>symplectic manifolds</i>
$J = [i]$	J is the complex structure on the Riemann sphere	$J = \begin{bmatrix} i & & \\ & i & \\ & & \ddots \end{bmatrix}$	not available in general!
$X_H := J\text{grad}(H)$	$X_H := J\text{grad}(H)$	$X_H := J\text{grad}(H)$	$X_H := -\omega^{-1}\text{d}H$ <i>notice the "-" sign!</i>
$X_H H = 0$	$X_H H = 0$	$X_H H = 0$	$X_H H = 0$
<i>area preserving</i> $\mathcal{L}_{X_H}\text{d}x \wedge \text{d}y = 0$	<i>area preserving</i> $\mathcal{L}_{X_H}\lambda_{\text{area}} = 0$	volume preserving, but preserves more...	<i>"k-dimensional area preserving"</i> $\mathcal{L}_{X_H}\omega^k = 0$

$\mathbb{C} = \mathbb{R}^2$	S^2	$\mathbb{C}^n = \mathbb{R}^{2n}$	(M, ω) <i>symplectic manifolds</i>
Converse? True for $\mathbb{R}^2$ but not true for general <i>open subsets!</i>	Converse? true for S^2 !	Converse for volume preservation is false for $n > 2$, even for linear flows!	Converse is false for $\dim M > 2$! However, it is true that vector fields that preserve the <i>symplectic 2-form</i> are Hamiltonian on a simply connected M .
Linear maps that are <i>generated</i> by Hamiltonian vector fields are just the area preserving ones $Sp_{\mathbb{R}}(2) = SL_{\mathbb{R}}(2)$	What could be "linear" flows? well isometries $SO_{\mathbb{R}}(3)$ preserve area, so we have one nice Lie group action on the 2-sphere.	$SL_{\mathbb{R}}(2n) \supset Sp_{\mathbb{R}}(2n) \supset U(n)$	symplectic G -actions, Hamiltonian G -actions

diagonal $T^2 \curvearrowright \mathbb{C}^2$

AKA the dynamics of 2 harmonic oscillators

On the symplectic 4-manifold

$$\left(\mathbb{R}^4, \sum_{i=1}^2 dp_i \wedge dq_i \right) = \left(\mathbb{C}^2, \frac{1}{2i} \sum_{i=1}^2 dz_i \wedge d\bar{z}_i \right)$$

we have the following Hamiltonian vector field and flow

equations	vector field	flow	Hamiltonian
$\begin{aligned}\dot{p}_1 &= -a_1 q_1 \\ \dot{q}_1 &= a_1 p_1 \\ \dot{p}_2 &= -a_2 q_2 \\ \dot{q}_2 &= a_2 p_2\end{aligned}$ or in more revealing, matrix form $\begin{bmatrix} \dot{p}_1 \\ \dot{q}_1 \\ \dot{p}_2 \\ \dot{q}_2 \end{bmatrix} = \begin{bmatrix} & -a_1 & & \\ a_1 & & & \\ & & & -a_2 \\ & & a_2 & \end{bmatrix} \begin{bmatrix} p_1 \\ q_1 \\ p_2 \\ q_2 \end{bmatrix}$ where the matrix is a block matrix with a scaled $\pi/2$-rotation matrix	$\omega_1 \underbrace{\left(-q_1 \frac{\partial}{\partial p_1} + p_1 \frac{\partial}{\partial q_1} \right)}_{X_1} +$		$\omega_1 \underbrace{\frac{p_1^2 + q_1^2}{2}}_{H_1} + \omega_2 \underbrace{\frac{p_2^2 + q_2^2}{2}}_{H_2}$

equations	vector field	flow	Hamiltonian
$\begin{aligned}\dot{z}_1 &= i\omega_1 z_1 \\ \dot{z}_2 &= i\omega_2 z_2\end{aligned}$ or $\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \end{bmatrix} = \begin{bmatrix} i\omega_1 & \\ & i\omega_2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$	$\underbrace{\omega_1 i z_1 \frac{\partial}{\partial z_1}}_{X_1} + \underbrace{\omega_2 i z_2 \frac{\partial}{\partial z_2}}_{X_2}$ $= \omega_1 X_1 + \omega_2 X_2$	$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \xrightarrow{t} \begin{bmatrix} \exp(i\omega_1 t) z_1 \\ \exp(i\omega_2 t) z_2 \end{bmatrix}$	$\omega_1 \underbrace{\frac{z_1 \bar{z}_1}{2}}_{H_1} + \omega_2 \underbrace{\frac{z_2 \bar{z}_2}{2}}_{H_2}$

We may rephrase in terms of the diagonal action on $\mathbb{C}^2$ by $\mathbb{T}^2$

action	generating vector field	moment map
diagonal $\mathbb{T}^2$-action on $\mathbb{C}^2$ <i>with both frequencies = 1</i> given by $\mathbb{T}^2 \curvearrowright \mathbb{C}^2$ $\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \xrightarrow{t_1, t_2} \begin{bmatrix} \exp(it_1)z_1 \\ \exp(it_2)z_2 \end{bmatrix}$	the generators of the $\mathbb{T}^2$-action are the vector fields X_1, X_2 with $[X_1, X_2] = 0$	the H_1, H_2 above with $\{H_1, H_2\} = 0$ give with moment map $(H_1, H_2) : \mathbb{C}^2 \rightarrow \mathbb{R}^2$ $(z, z_2) \mapsto \frac{1}{2}(z_1\bar{z}_1, z_2\bar{z}_2)$ of the Hamiltonian $\mathbb{T}^2$-action
from the homomorphism $S^1 \rightarrow \mathbb{T}^2 \curvearrowright \mathbb{C}^2$ $t \mapsto l_1 t + l_2 t$ for $l_1, l_2 \in \mathbb{Z}$ we get a S^1-action on $\mathbb{C}^2$ with <i>integer frequencies</i>	the generator is $l_1 X_1 + l_2 X_2$	
$\mathbb{R} \rightarrow \mathbb{T}^2$	$\omega_1 X_1 + \omega_2 X_2$	
$\mathbb{T}^2 \rightarrow \mathbb{T}^2$		

restricting to the invariant 3-sphere

We restrict to the 3-sphere

$$S_h^3 = \{(p_1, q_1, p_2, q_2) \in \mathbb{R}^4 | p_1^2 + q_1^2 + p_2^2 + q_2^2 = 2h\} \cong_{\text{Man}} S^3$$

which is invariant under the flow of $\omega_1 X_1 + \omega_2 X_2$ and has orbits of $H_1 = h$. On this sphere, we have the invariant tori

$$T_{h_1, h_2}^2 := \{(p_1, q_1, p_2, q_2) \in \mathbb{R}^4 | p_1^2 + q_1^2 = 2h_1, p_2^2 + q_2^2 = 2h_2\}$$

with $\omega_1 h_1 + \omega_2 h_2 = h$.

The projection

$$S^3 \setminus \{(0, 0, 0, 1)\} \rightarrow \mathbb{R}^3$$
$$(x, y, z, w) \mapsto (X, Y, Z) := \frac{1}{1-w}(x, y, z)$$

maps

- the orbit

$$(p_1, q_1, 0, 0) \mapsto (p_1, q_1, 0)$$

where $p_1^2 + q_1^2 = 2h_1$, the image is thus a circle in the x-y plane in $\mathbb{R}^3$

- the orbit

$$(0, 0, p_2, q_2) \mapsto \left(0, 0, \frac{p_2}{1 - q_2}\right)$$

whose image is the z-axis

- $$T_{h_1, h_2}^2 \mapsto \left\{ (X, Y, Z) \left| X^2 + Y^2 = \frac{2h_1}{(1 - w)^2}, Z^2 = \frac{2h_2 - w^2}{(1 - w)^2}, w \neq 1 \right. \right\}$$

which should be a torus in $\mathbb{R}^3$

-

restricting to the invariant tori: a torus translation

The subset

$$\begin{aligned} T_{a,b}^2 &:= \{(p_1, q_1, p_2, q_2) \in \mathbb{R}^4 | p_1^2 + q_1^2 = 2a, p_2^2 + q_2^2 = 2b\} \subseteq (\mathbb{R}^2 \setminus \{0\})^2 \\ S_h^3 &:= \{(p_1, q_1, p_2, q_2) \in \mathbb{R}^4 | p_1^2 + q_1^2 + p_2^2 + q_2^2 = 2h\} \subseteq \mathbb{R}^4 \setminus \{0\} \end{aligned}$$

which is preserved by the flow of $\omega_1 X_1 + \omega_2 X_2$.

Thus $(\mathbb{R}^2 \setminus \{0\})^2$ has a torus fibration



coordinates	X_1	$\omega_1 X_1 + \omega_2 X_2$	flow
On $\mathbb{R}^4$ we have (p_1, q_1, p_2, q_2)	$-q_1 \frac{\partial}{\partial p_1} + p_1 \frac{\partial}{\partial q_1}$	$\omega_1 \underbrace{\left(-q_1 \frac{\partial}{\partial p_1} + p_1 \frac{\partial}{\partial q_1} \right)}_{X_1} +$	
On $(\mathbb{R}^2 \setminus \{0\})^2$ we have the coordinates $r_1^2 = p_1^2 + q_1^2$ θ_1 $r_2^2 = p_2^2 + q_2^2$ θ_2	$\frac{\partial}{\partial \theta_1}$	$\omega_1 \frac{\partial}{\partial \theta_1} + \omega_2 \frac{\partial}{\partial \theta_2}$ which is a torus translation $T_{a,b}^2$	

Flow of a general vector field

$$\omega_1 \frac{\partial}{\partial \theta_1} + \omega_2 \frac{\partial}{\partial \theta_2}$$

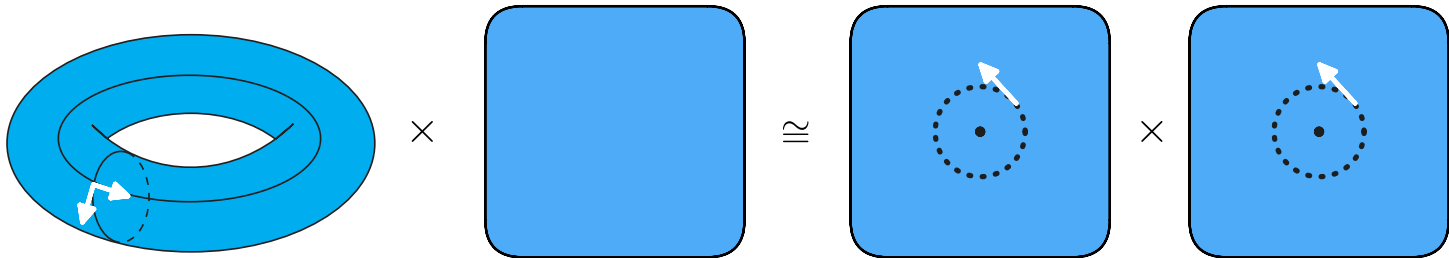
is given by

$$\exp \left(t \left(\omega_1 \frac{\partial}{\partial \theta_1} + \omega_2 \frac{\partial}{\partial \theta_2} \right) \right) \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} + t \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix}$$

whose flows have properties

if	flow is	which gives a	that is coming from
$\omega_1/\omega_2 \in \mathbb{Q}$	periodic	S^1 -action on T^2	composition $S^1 \hookrightarrow T^2 \curvearrowright T^2$
$\omega_1/\omega_2 \in \mathbb{R} \setminus \mathbb{Q}$	not periodic, minimal (orbits are dense)	$\mathbb{R}$ -action on T^2	composition $\mathbb{R} \hookrightarrow T^2 \curvearrowright T^2$

decomposition into torus translation $\times$ trivial



$$\frac{\partial}{\partial \theta_1}$$

$$0$$

$$\mapsto$$

$$-q_1 \frac{\partial}{\partial p_1} + p_1 \frac{\partial}{\partial q_1}$$

$$0$$

$$\frac{\partial}{\partial \theta_2}$$

$$0$$

$$\mapsto$$

$$0$$

$$-q_2 \frac{\partial}{\partial p_2} + p_2 \frac{\partial}{\partial q_2}$$

$$T^2 \hookrightarrow T^2 \times \mathbb{R}^2 \cong_{T^2} (\mathbb{R}^2 \setminus \{0\})^2$$

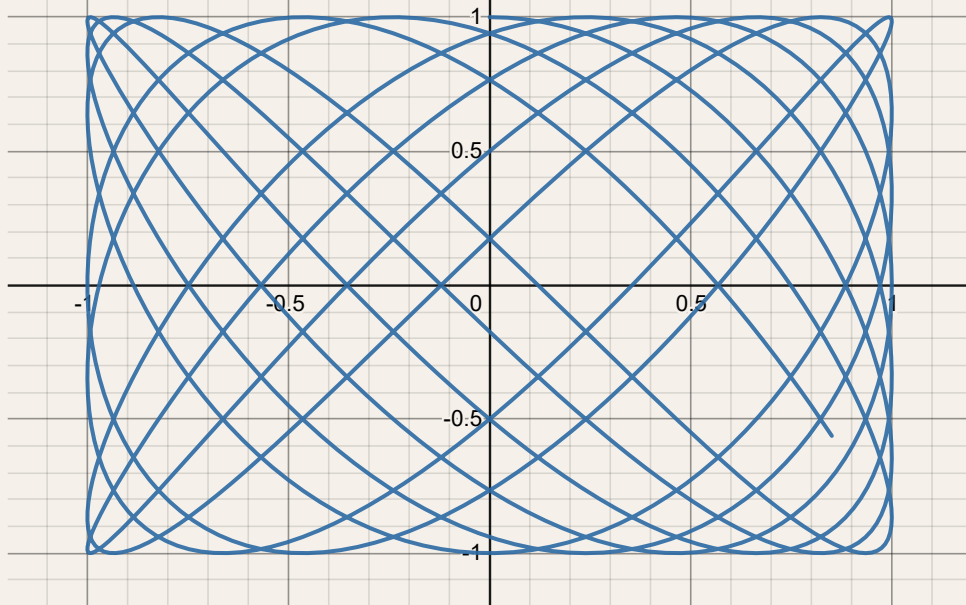
$$v_1 \mapsto \frac{\partial}{\partial \theta_1} \times 0 \mapsto X_1$$

$$v_2 \mapsto \frac{\partial}{\partial \theta_2} \times 0 \mapsto X_2$$

projecting onto (q_1, q_2)



$$\{(\sin 45t, \cos 65t) | t \in [0, 1]\}$$



Projecting $(0, q_1, 0, q_2)$ we have

$$(q_1, q_2) \xrightarrow{t} (\sin(\omega_1 t)q_1, \sin(\omega_2 t)q_2)$$

[1]

rational oscillator when $\omega_1, \omega_2 \in \mathbb{Z}$

A $S^1 \cong U(1)$ -action (or representation) on $\mathbb{R}^4 = \mathbb{C}^2$

irrational oscillator when $\omega_1/\omega_2 \in \mathbb{R} \setminus \mathbb{Q}$

as a Lax pair

as a Lax pair

We define

$$L := \begin{bmatrix} p & q \\ q & -p \end{bmatrix}$$
$$M := \frac{\omega}{2} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

then

$$\begin{aligned} [M, L] &= \frac{\omega}{2} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} p & q \\ q & -p \end{bmatrix} - \begin{bmatrix} p & q \\ q & -p \end{bmatrix} \frac{\omega}{2} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \\ &= \frac{\omega}{2} \begin{bmatrix} -q & p \\ p & q \end{bmatrix} - \frac{\omega}{2} \begin{bmatrix} q & -p \\ -p & -q \end{bmatrix} \\ &= \omega \begin{bmatrix} -q & p \\ p & q \end{bmatrix} \end{aligned}$$

Hence, the oscillator is equivalent to

$$\dot{L} = [M, L]$$

thus giving us conserved quantities

$$\mathrm{tr}(L^2) = \mathrm{tr}\begin{pmatrix} p^2 + q^2 & ? \\ ? & 0 \end{pmatrix} = p^2 + q^2$$

$$L := \begin{bmatrix} p_1 & q_1 \\ q_1 & -p_1 \end{bmatrix} + \begin{bmatrix} p_2 & q_2 \\ q_2 & -p_2 \end{bmatrix} z$$

$$M := \frac{\omega}{2} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$[M, A_1 + A_2 z] = [M, A_1] + [M, A_2] z$$

then

$$\dot{L} = [M, L]$$

should give the equation for isotropic harmonic oscillator.

Now

$$\begin{aligned} yI_2 - A_1 - A_2 z &= \begin{bmatrix} y - p_1 - zp_2 & -q_1 - zq_2 \\ -q_1 - zq_2 & y + p_1 + zp_2 \end{bmatrix} \\ \det(yI_2 - A_1 - A_2 z) &= y^2 - (p_1 + zp_2)^2 - (q_1 + zq_2)^2 \\ &= y^2 - (p_1^2 + q_1^2) - (p_2^2 + q_2^2)z^2 - 2z(p_1p_2 + q_1q_2) \end{aligned}$$

which is a conic section over which

$$(y, z) \mapsto \ker(yI_2 - A(z))$$

defines a line bundle.

[2]

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1. <https://www.desmos.com/calculator/irnyqunfsm> ↩
 2. <https://www.desmos.com/calculator/2n2puejwvk> ↩

Hamiltonian flows on symplectic manifold

We want a smooth tensor

$$\omega : T^*M \rightarrow TM$$

that is, smoothly varying (on $p \in M$) linear maps

$$\omega_p : T_p^*M \rightarrow T_p^*M$$

with which we want to define

$$X_H := -\omega^{-1}dH \iff dH(-) = -\omega(X_H, -)$$

as the Hamiltonian vector field of H such that

- **flow of X_H must preserve H**

$$0 = X_H H = dH(X_H) = -\omega(X_H, X_H)$$

$\iff \omega$ is **alternating**, so a 2-form

- and **flow of X_H must preserve ω**

$$0 = \mathcal{L}_{X_H}\omega = \iota_{X_H}d\omega + \underbrace{d\iota_{X_H}\omega}_{-dH}$$

which is true when $d\omega = 0$

Thus

📌 **Definition.** A symplectic manifold (M, ω) is a smooth manifold M with a closed non-degenerate 2-form ω .

properties of the symplectic form

Proposition: ω^n defines a volume form

Darboux local coordinates on symplectic manifolds

The Darboux Theorem

Our next theorem is one of the most fundamental results in symplectic geometry. It is a nonlinear analogue of the canonical form for a symplectic tensor given in Proposition 22.7. It illustrates the most dramatic difference between symplectic structures and Riemannian metrics: unlike the Riemannian case, there is no local obstruction to a symplectic structure being locally equivalent to the standard flat model.

Theorem 22.13 (Darboux). *Let (M, ω) be a $2n$ -dimensional symplectic manifold. For any $p \in M$, there are smooth coordinates $(x^1, \dots, x^n, y^1, \dots, y^n)$ centered at p in which ω has the coordinate representation*

$$\omega = \sum_{i=1}^n dx^i \wedge dy^i. \quad (22.5)$$

We will prove the theorem below. Any coordinates satisfying (22.5) theorem are called *Darboux coordinates*, *symplectic coordinates*, or *canonical coordinates*.

Proof of the Darboux theorem. Let ω_0 denote the given symplectic form on M , and let $p_0 \in M$ be arbitrary. The theorem will be proved if we can find a smooth coordinate chart (U_0, φ) centered at p_0 such that $\varphi^* \omega_1 = \omega_0$, where $\omega_1 = \sum_{i=1}^n dx^i \wedge dy^i$ is the standard symplectic form on $\mathbb{R}^{2n}$. Since this is a local question, by choosing smooth coordinates $(x^1, \dots, x^n, y^1, \dots, y^n)$ in a neighborhood of p_0 , we may replace M with an open ball $U \subseteq \mathbb{R}^{2n}$. Proposition 22.7 shows that we can arrange by a linear change of coordinates that $\omega_0|_{p_0} = \omega_1|_{p_0}$.

Let $\eta = \omega_1 - \omega_0$. Because η is closed, the Poincaré lemma (Theorem 17.14) shows that we can find a smooth 1-form α on U such that $d\alpha = -\eta$. By subtracting a constant-coefficient (and thus closed) 1-form from α , we may assume without loss of generality that $\alpha_{p_0} = 0$.

For each $t \in \mathbb{R}$, define a closed 2-form ω_t on U by

$$\omega_t = \omega_0 + t\eta = (1-t)\omega_0 + t\omega_1.$$

Let J be a bounded open interval containing $[0, 1]$. Because $\omega_t|_{p_0} = \omega_0|_{p_0}$ is nondegenerate for all t , a simple compactness argument shows that there is some neighborhood $U_1 \subseteq U$ of p_0 such that ω_t is nondegenerate on U_1 for all $t \in \bar{J}$. Because of

this nondegeneracy, the smooth bundle homomorphism $\widehat{\omega}_t: TU_1 \rightarrow T^*U_1$ defined by $\widehat{\omega}_t(X) = X \lrcorner \omega_t$ is an isomorphism for each $t \in \bar{J}$.

Define a smooth time-dependent vector field $V: J \times U_1 \rightarrow TU_1$ by $V_t = \widehat{\omega}_t^{-1}\alpha$, or equivalently

$$V_t \lrcorner \omega_t = \alpha.$$

Our assumption that $\alpha_{p_0} = 0$ implies that $V_t|_{p_0} = 0$ for each t . If $\theta: \mathcal{E} \rightarrow U_1$ denotes the time-dependent flow of V , it follows that $\theta(t, 0, p_0) = p_0$ for all $t \in J$, so $J \times \{0\} \times \{p_0\} \subseteq \mathcal{E}$. Because $\mathcal{E}$ is open in $J \times J \times M$ and $[0, 1]$ is compact, there is some neighborhood U_0 of p_0 such that $[0, 1] \times \{0\} \times U_0 \subseteq \mathcal{E}$.

For each $t_1 \in [0, 1]$, it follows from Proposition 22.15 that

$$\begin{aligned} \frac{d}{dt} \Big|_{t=t_1} (\theta_{t,0}^* \omega_t) &= \theta_{t_1,0}^* \left(\mathcal{L}_{V_{t_1}} \omega_{t_1} + \frac{d}{dt} \Big|_{t=t_1} \omega_t \right) \\ &= \theta_{t_1,0}^* (V_{t_1} \lrcorner d\omega_{t_1} + d(V_{t_1} \lrcorner \omega_{t_1}) + \eta) \\ &= \theta_{t_1,0}^* (0 + d\alpha + \eta) = 0. \end{aligned}$$

Therefore, $\theta_{t,0}^* \omega_t = \theta_{0,0}^* \omega_0 = \omega_0$ for all t . In particular, $\theta_{1,0}^* \omega_1 = \omega_0$. It follows from Theorem 9.48(c) that $\theta_{1,0}$ is a diffeomorphism onto its image, so it is a coordinate map. Because $\theta_{1,0}(p_0) = p_0 = 0$, these coordinate are centered at p_0 . $\square$

Moser's theorem on volumes

Theorem 2. (Moser) Assume M is a compact, connected and oriented manifold of dimension m without boundary. If α and β are two volume forms such that their total volumes agree, i.e.,

$$(1.40) \quad \int_M \alpha = \int_M \beta,$$

then there is a diffeomorphism u of M satisfying $u^*\beta = \alpha$.

Consequently the total volume is the only invariant of volume-preserving diffeomorphisms.

Proof. We proceed as in the proof of Darboux's theorem and deform the volume form α into β defining

$$\alpha_t = (1 - t)\alpha + t\beta, \quad 0 \leq t \leq 1.$$

These forms α_t are volume forms since locally α and β are represented by $\alpha = a(x)dx_1 \wedge \dots \wedge dx_m$ and $\beta = b(x)dx_1 \wedge \dots \wedge dx_m$ with nonvanishing smooth functions a and b , which, by assumption (1.40), must have the same sign. We shall construct a family φ^t of diffeomorphisms satisfying

$$(1.41) \quad \left(\varphi^t\right)^* \alpha_t = \alpha, \quad \varphi^0 = id$$

for $0 \leq t \leq 1$, so that the diffeomorphism $u = \varphi^1$ will solve our problem. Since M is compact, connected and oriented we conclude from $\int_M (\beta - \alpha) = 0$ that

$$(1.42) \quad \beta - \alpha = d\gamma$$

for some $(m - 1)$ -form γ on M . This is a special case of the de Rham theorem. Since α_t is a volume form we find a unique time-dependent vector field X_t on M solving the linear equation

$$i_{X_t} \alpha_t = -\gamma$$

for $0 \leq t \leq 1$. Denote by φ^t the flow of this vector field X_t satisfying $\varphi^0 = id$. Since M is compact it exists for all t . Since $d\alpha_t = 0$ for volume forms we find, again using Cartan's formula,

$$\begin{aligned} \frac{d}{dt} \left(\varphi^t \right)^* \alpha_t &= \left(\varphi^t \right)^* \left(L_{X_t} \alpha_t + \frac{d}{dt} \alpha_t \right) \\ &= \left(\varphi^t \right)^* \left(d(i_{X_t} \alpha_t) + \beta - \alpha \right), \end{aligned}$$

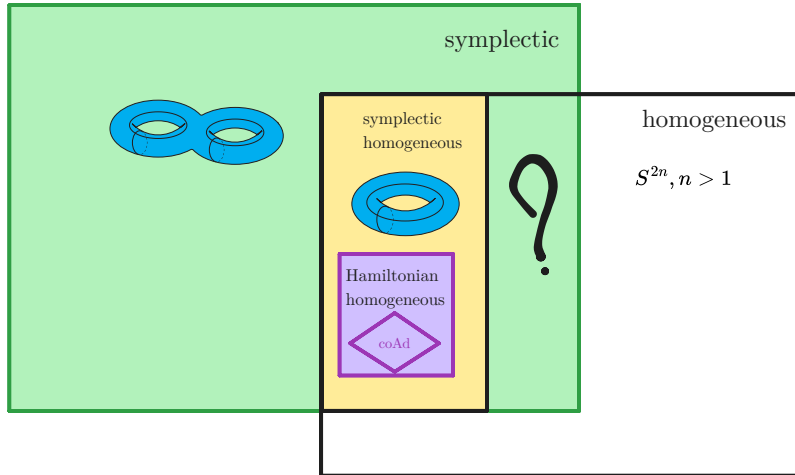
which vanishes since $d(i_{X_t} \alpha_t) + \beta - \alpha = d(i_{X_t} \alpha_t + \gamma) = 0$ by our choice of the vector field X_t . Therefore (1.41) holds and the proof is finished. ■

references

- Jack Lee's book on smooth manifolds
- <http://staff.ustc.edu.cn/~wangzuoq/Courses/15S-Symp/SympGeom.html>
- Ana Canas da Silver, Lectures on Symplectic Geometry, Lecture Notes in Mathematics 1764, Springer.

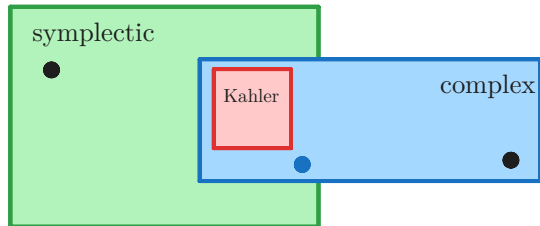
Symplectic manifold

📌 **Definition.** A symplectic manifold (M, ω) is a smooth manifold M with a closed non-degenerate 2-form ω .



smooth even dimensional orientable manifolds

almost symplectic = almost complex
= almost Hermitian



Manifolds with compatible triples

Definition. Smooth manifold with compatible triples

A smooth manifold with a compatible triple

$$(M, \omega, J, g)$$

consists of

- almost symplectic structure ω
- almost complex structure J
- a Riemannian metric g

which satisfy

$$J = g^{-1}\omega^{-1}$$

This gives

- two volume forms λ_g and $\omega^{\dim M/2}$, which *may* differ but of course there exists a never vanishing $s \in \mathcal{C}^\infty(M)$ such that

$$s\lambda_g = \omega^{\dim M/2}$$

- J -rotation of gradient vector fields **match** with ω -Hamiltonian vector fields

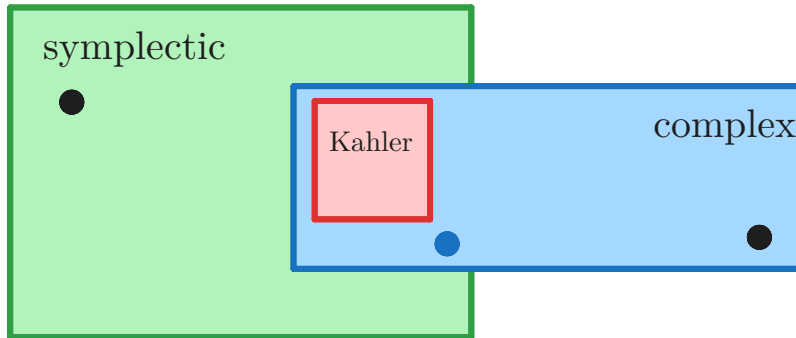
$$\begin{aligned}
 g(\text{grad}(f), -) &= df(-) = -\omega(X_f, -) \\
 \implies g(\text{grad}(f), -) &= -g(JX_f, -) \\
 \implies X_f &= J\text{grad}(f)
 \end{aligned}$$

Proposition:

given (ω almost symplectic)	with the condition	we have the third structure
ω, J	$\omega(-, J-)$ is a Riemannian metric	$g := \omega(-, J-)$ is Riemannian
ω, g	none	J using polar decomposition
g, J	J is orthogonal, as in $g(J-, J-) = g(-, -)$ $\iff g(J-, -) = -g(-, J-)$ (almost Hermitian)	$\omega := g(J-, -)$ is almost symplectic

smooth even dimensional orientable manifolds

almost symplectic = almost complex
= almost Hermitian



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1. [S Ramanan - Global Calculus-American Mathematical Society \(2005\), p.193](#) ↩

Orbits of smooth Lie group actions

Let $\theta : G \curvearrowright M$ be a smooth left Lie group action on a smooth manifold M and $p \in M$. Then the orbit map

$$\begin{aligned}\theta_{(p)} : G &\rightarrow M \\ g &\mapsto gp\end{aligned}$$

is smooth and has constant rank, so the isotropy group $G_p = \theta_{(p)}^{-1}(p)$ is a **properly embedded Lie subgroup** of G . Thus

$$\theta_{(p)} : \frac{G}{G_p} \rightarrow M$$

is an injective immersion making the image, that is, the orbit $G \{p\}$ is an immersed submanifold

$$\iota : (G \{p\}, \theta_{(p)*}\mathfrak{T}_{G/G_p}, \theta_{(p)*}\mathcal{A}_{G/G_p}) \hookrightarrow M$$

which is embedded if the action is proper.

Proposition 7.26 (Properties of the Orbit Map). *Suppose θ is a smooth left action of a Lie group G on a smooth manifold M . For each $p \in M$, the orbit map $\theta^{(p)}: G \rightarrow M$ is smooth and has constant rank, so the isotropy group $G_p = (\theta^{(p)})^{-1}(p)$ is a properly embedded Lie subgroup of G . If $G_p = \{e\}$, then $\theta^{(p)}$ is an injective smooth immersion, so the orbit $G \cdot p$ is an immersed submanifold of M .*

21-17. Suppose a Lie group acts smoothly (but not necessarily properly or freely) on a smooth manifold M . Show that each orbit is an immersed submanifold of M , which is embedded if the action is proper.

Ad^* orbits are symplectic manifolds

📌 **Definition.** Symplectic structure on coadjoint orbits of Lie groups

Let

$$\Omega \subset \mathfrak{g}^*$$

be a coadjoint orbit $\text{Ad}^* : G \curvearrowright \mathfrak{g}^*$ of a Lie group G . Let $F \in \Omega$.

- By definition Ω is a transitive (left) G -space with stabilizer of F being $\text{Ad}_{G,F}^*$ (a closed subgroup).
- Thus the surjective smooth map

$$\begin{aligned} G &\rightarrow \Omega \\ g &\mapsto \text{Ad}_g^*(F) \end{aligned}$$

with π descends to bijection

$$\frac{G}{\text{Ad}_{G,F}^*} \rightarrow \Omega$$

giving a smooth structure on the orbit Ω making it a *smooth immersed manifold* in $\mathfrak{g}^*$.

We construct a anti-symmetric bilinear map

$$\omega_F : \frac{\mathfrak{g}}{\text{Lie}(\text{Ad}_{G,F}^*)} \times \frac{\mathfrak{g}}{\text{Lie}(\text{Ad}_{G,F}^*)} \rightarrow \mathbb{R}$$

$$(X, Y) \mapsto F([X, Y])$$

Proposition:

- The bilinear map

$$\omega_F : \frac{\mathfrak{g}}{\text{Lie}(\text{stab}_G(F))} \times \frac{\mathfrak{g}}{\text{Lie}(\text{stab}_G(F))} \rightarrow \mathbb{R}$$

$$(X, Y) \mapsto F([X, Y])$$

defines a Ad^* -invariant **symplectic form** on the orbit $\Omega \subset \mathfrak{g}$.

- The coadjoint action $G \curvearrowright \Omega$ is **Hamiltonian** with moment map

$$\iota : \Omega \hookrightarrow \mathfrak{g}^*$$

[1]

Thus

$$\text{Ad}^* : G \curvearrowright (\Omega, \omega) \subset \mathfrak{g}^*$$

is a **homogeneous Hamiltonian G -manifold**.

[2]

[3]

[4]

Lie group	coadjoint orbits
$SO(3, \mathbb{R}), SU(2, \mathbb{C})$	$\{0\}, S^2$
compact	Kahler ^[1-1]
nilpotent	?

1. <https://arxiv.org/pdf/0906.4915> ↩ ↪
2. <studenttheses.uu.nl/bitstream/handle/20.500.12932/7207/Maes%2CJMA2011.pdf?sequence=1&isAllowed=y#page=40.00> ↩
3. [https://people.math.sc.edu/kellerlv/Symplectic_Geometry_HW1%20\(41\).pdf](https://people.math.sc.edu/kellerlv/Symplectic_Geometry_HW1%20(41).pdf) ↩
4. <https://www.cambridge.org/core/journals/canadian-mathematical-bulletin/article/every-symplectic-manifold-is-a-linear-coadjoint-orbit/60EEA4D0397E72FAE554ADEE9EC800B0> ↩

Symplectic homogeneous manifolds

📌 **Definition.** A **symplectic homogeneous manifold** is a symplectic manifold with a transitive symplectic Lie group action.

[1]

- Ad^* orbits are symplectic manifolds

classification of homogeneous symplectic manifolds

Let X, G be simply connected and

$$G \curvearrowright (X, \omega)$$

by a homogeneous symplectic G -manifold. Thus it is a **almost Hamiltonian G -action**.

Now define

$$\bar{\mathfrak{g}} := \{(h, Z) \in \mathcal{C}^\infty(X) \times \mathfrak{g} \mid Z = -\omega^{-1}dh\}$$

with bracket

$$[(h_1, Z_1), (h_2, Z_2)] = [\{h_1, h_2\}, [Z_1, Z_2]]$$

then we have the exact sequence

$$\begin{array}{ccccccc} 0 & \rightarrow & \mathbb{R} & \rightarrow & \overline{\mathfrak{g}} & \rightarrow & \mathfrak{g} \rightarrow 0 \\ & & & & (h, Z) & \mapsto & Z \end{array}$$

The corresponding Lie group $\overline{G}$ we have

$$\overline{G} \rightarrow G \curvearrowright X$$

- Now define

$$\begin{array}{l} \Phi : X \rightarrow \overline{\mathfrak{g}}^* \\ x \mapsto \Phi_x : \Phi_x(h, Z) = h(x) \end{array}$$

- Then

$$\begin{array}{l} \omega(Z, -) = -dh \\ \dots \end{array}$$

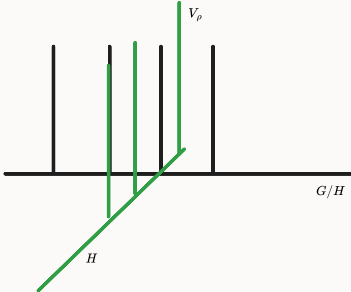
so this Φ is indeed a moment map.

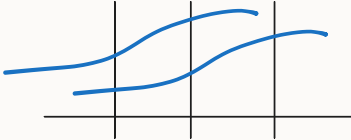
- By construction ...

[2]

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1. [Symplectic homogeneous space - Encyclopedia of Mathematics](#) ↩
 2. [https://people.math.sc.edu/kellerlv/Symplectic_Geometry_HW1%20\(41\).pdf](https://people.math.sc.edu/kellerlv/Symplectic_Geometry_HW1%20(41).pdf) ↩

Ind reps

	finite case	"H-sections"	sections on the fibred product bundle
		$ \begin{array}{c} G \times V_\rho \\ \downarrow \\ G \end{array} $ 	$ \begin{array}{ccc} G \curvearrowright & G \times_H V_\rho & \\ & \downarrow & \\ G \curvearrowright & G/H & \end{array} $
Ind_H^G	$V_\pi := \bigoplus_{G/H} V_\rho$	$= \{ f : G \rightarrow V_\rho \mid f(gh) = \rho(g)f(h) \}$	$\Gamma(G/H, G \times_H V_\rho)$ which are L^2 "a easier version of direct integrals"

	finite case	"H-sections"	sections on the fibred product bundle
G -action	$\pi(gg_i)(v,i) = (\rho(h)v,j)$	$(\pi(g)f)(g) = f(g^{-1}g)$	$(gs)(g_1H) = gs(g^{-1}g_1H)$ 
inner product on the bundle			$\langle [g,v],[g,w]\rangle = \langle v,w\rangle_{V_\rho}$ (which is well-defined)
Let ν be a G -invariant measure on G/H (not available in general)		$\langle f_1,f_2\rangle := \int_{G/H} \langle f_1(g),f_2(g) \rangle \nu(gH)$	$\langle s_1,s_2\rangle = \int_{G/H} \langle s_1(gH),s_2(gH) \rangle \nu(gH)$

Proposition:

$$\Gamma(G/H,G\times_H V_\rho) = \Gamma_H(G,G\times V_\rho)$$



$$f \mapsto \left(gH \mapsto [(g,f(h))]\right)$$

- and

$$f_s \leftarrow s$$

where f_s is defined to be the element

$$s(gH) = [(g, f_s(g))]$$

Proposition: The induced representation is unitary.



$$\begin{aligned} \langle \pi(g)f_1, \pi(g)f_2 \rangle &= \int_{G/H} \langle f_1(g^{-1}g_1), f_2(g^{-1}g_1) \rangle d\nu_{g_1H} \\ &= \int_{G/H} \langle f_1(g_1), f_2(g_1) \rangle d\nu_{g_1H} \end{aligned}$$

examples

-

$$\text{Ind}_H^G(1) \cong L^2(G/H)$$

Koopman representations

- Let

$$G \curvearrowright (X, \mathfrak{M}_X, \mu)$$

be measure preserving group action. Then it induces a **unitary representation**

$$G \curvearrowright L^2(X)$$

- Let

$$G \curvearrowright (X, \mathfrak{M}_X, \mu)$$

be measure *class* preserving group action.

- Then we define

$$G \curvearrowright L^2(X)$$

$$(\pi(g)f)(x) = f(g^{-1}x) \sqrt{\frac{dg_*\mu}{d\mu}}(x)$$

to get a **unitary representation**.

- To be well-defined we must check the cocycle condition

$$\pi(g_1 g_2) f = \pi(g_1) \pi(g_2) f \iff \frac{d(g_1 g_2)_* \mu}{d\mu}(x) = \frac{dg_{1*} \mu}{d\mu}(x) \frac{dg_{2*} \mu}{d\mu}(g_1^{-1} x)$$

which is true!

•

$$L^2(X, V)$$